

System Analysis & Design Math

A very basic cash flow projection for an IT Projects is, in the Year 0,1,2,3 costs are 50000, 8000, 10000, 14000 and benefits are 0, 25000, 30000, 48000. Find out the followings for the above conditions:

- (i) Return on Investment
- (ii) Break-Even Point

Given (years 0,1,2,3)

Costs: 50,000 ; 8,000 ; 10,000 ; 14,000 → **Total costs = 82,000.**

Benefits: 0 ; 25,000 ; 30,000 ; 48,000 → **Total benefits = 103,000.**

Net (total) = Total benefits – Total costs = 103,000 – 82,000 = 21,000.

(i) Return on Investment (simple ROI)

Using the usual simple ROI = (Net gain) / (Total cost):

$$ROI = \frac{21,000}{82,000} = \frac{21}{82} \approx 0.25610 \approx 25.61\%$$

(If you instead define ROI relative to the initial investment (year-0 cost = 50,000), then ROI = 21,000 / 50,000 = 42%. Specify which definition you want if different.)

(ii) Break-Even Point (when cumulative cash ≥ 0)

Compute cumulative cash each year (benefits – costs, cumulative):

- End of year 0: 0 - 50,000 = -50,000 (cum = -50,000)
- End of year 1: 25,000 - 8,000 = +17,000 → cum = -33,000
- End of year 2: 30,000 - 10,000 = +20,000 → cum = -13,000
- End of year 3: 48,000 - 14,000 = +34,000 → cum = +21,000

Cumulative turns from -13,000 (end of year 2) to +21,000 (end of year 3), so break-even occurs during year 3.

Fraction of year 3 needed = amount to cover / year-3 net = 13,000 / 34,000 = 0.38235...

So break-even $\approx$ **2.38 years** after start (i.e., about 0.38 of the way through year 3).

If you prefer a whole-year answer: **break-even in year 3.**